AIM

To write a C program to perform various operations on an array of size 10 such as counting the number of even elements, sorting the array in ascending and descending order, and removing duplicate elements.

PROCEDURE

1. Start the program.
2. Declare an integer array of size 10 and required variables.
3. Read 10 elements from the user and store them in the array.
4. Traverse the array and count the number of even elements.
5. Sort the array in ascending order using nested for loops.
6. Display the sorted array in ascending order.
7. Sort the array in descending order using nested for loops.
8. Display the sorted array in descending order.
9. Compare array elements and remove duplicate elements by shifting values.
10. Display the array after removing duplicate elements.
11. End the program.

CODE

```
/*
File Name   : array.c
Author      : Monish Kanna
Date        : 12-01-2026
Purpose     : Program to perform array operations
              (count even numbers, sort ascending,
               sort descending, remove duplicates)
*/
#include <stdio.h>
int main()
{
    int arr[10], i, j, k, temp;
    int evenCount = 0;
```

```
int n = 10;
printf("Enter 10 array elements:\n");
for (i = 0; i < n; i++)
{ scanf("%d", &arr[i]);   }
for (i = 0; i < n; i++)
{
    if (arr[i] % 2 == 0)
    { evenCount++;  }
}
printf("\nNumber of even elements = %d\n", evenCount);
for (i = 0; i < n - 1; i++)
{
    for (j = i + 1; j < n; j++)
    {
        if (arr[i] > arr[j])
        {
            temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }
}
printf("\nArray in Ascending Order:\n");
for (i = 0; i < n; i++)
{    printf("%d ", arr[i]); }
for (i = 0; i < n - 1; i++)
{
    for (j = i + 1; j < n; j++)
    {
        if (arr[i] < arr[j])
        {
            temp = arr[i];
```

```
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }
    printf("\n\nArray in Descending Order:\n");
    for (i = 0; i < n; i++)
    { printf("%d ", arr[i]); }
for (i = 0; i < n; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (arr[i] == arr[j])
            {
                for (k = j; k < n - 1; k++)
                {   arr[k] = arr[k + 1]; }
                n--;
                j--;
            }
        }
    }
    printf("\n\nArray after removing duplicate elements:\n");
    for (i = 0; i < n; i++)
    { printf("%d ", arr[i]); }
    return 0;
}
```

OUTPUT

```
PROBLEMS    TERMINAL    OUTPUT    PORTS

PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week1_practice5\" ; if ($?)
{ gcc array.c -o array } ; if ($?) { .\array }
Enter 10 array elements:
1
5
9
2
6
3
4
7
8
0

Number of even elements = 5

Array in Ascending Order:
0 1 2 3 4 5 6 7 8 9

Array in Descending Order:
9 8 7 6 5 4 3 2 1 0

Array after removing duplicate elements:
9 8 7 6 5 4 3 2 1 0
```

RESULT

The C program was executed successfully.
The program correctly counted the number of even elements, sorted the array in ascending and descending order, and removed duplicate elements from the array.